

Michael Liu

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Computer Science (Machine Learning Concentration)

August 2025 – May 2028

- Selected Coursework: Principles of Imperative Computing, Functional Programming, Introduction to Computer Systems, Mathematical Foundations for Computer Science, Linear and Matrix Algebra, Vector Calculus

EXPERIENCE

Web Developer

September 2025 – Present

ScottyLabs @ Carnegie Mellon University

Pittsburgh, PA

- Built a star rating and user review system for cmueats.com, serving 10,000+ monthly users campus dining info.
- Led development of user reviews and location sorting by location, rating, and opening time; integrated dining REST API with React Query for caching, prefetching, and refetching.

Research Assistant

February 2026 – Present

Robotics Institute @ Carnegie Mellon University

Pittsburgh, PA

- Contributed to a real-time data collection codebase for dexterous robot manipulation research using ROS, stereo cameras, Apple Vision Pro hand tracking, and HDF5 trajectory storage.
- Collected human hand demonstration data and troubleshoot hardware/software issues across multi-camera setups, AprilTag calibration, and synchronized recording workflows.

Research Intern

June 2022 – August 2024

The University of Texas at Dallas

Dallas, TX

- Modeled an autonomous driving agent in CARLA using a variational autoencoder, imitation learning, and Soft Actor-Critic; achieved +68% distance traveled, improved lane adherence, and more consistent speeds.
- Trained PyTorch MLP, CNN, and LSTM regressors on NASA Li-ion datasets across varying temperatures to model SOH curves (battery life and degradation); determined LSTM to be most accurate (RMSE 0.02).
- Researched inductive logic programming in Prolog to generate interpretable rules from machine learning models.

Teaching Assistant

June 2022 – August 2024

Dr. Meng's Coding Camp

Dallas, TX

- Taught 100+ students grades 6 – 10 data structures/algorithms, Linux/Raspberry Pi, and introductory ML.
- Fostered STEM engagement by designing project-based curriculum, hosting office hours, and grading homework.

PROJECTS

LeapASL | Java, Python, TensorFlow/Keras, scikit-learn, Leap Motion Controller

February 2023

- Built a Java Swing GUI app to capture 31,000+ ASL letters from 6 subjects using a Leap Motion Controller.
- Trained and integrated an RNN and SVM with accuracies of 94% to construct, correct, and read text in real time.

VolunTEEN | Dart, Flutter, Google Firebase

November 2021

- Led a team of 4 to develop an app using Dart and Flutter with Google Firebase user authentication, location-based volunteer matching, and event organization to connect hosts with local volunteers; earned 2nd/200+ in hackathon.

Flaurapedia | Java, XML, Android Studio, TensorFlow Lite, Google Firebase

August 2021

- Developed an app built with Android Studio (Java/XML), TensorFlow Lite, and Google Firebase featuring user authentication, plant identification via image upload or capture, and community post sharing/browsing.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript/TypeScript, HTML/CSS, Dart, SML/NJ, Prolog

Frameworks: React, Unity, Flutter, PyTorch, TensorFlow, NumPy

Developer Tools: Git/GitHub, Linux, Bash, Google Firebase, Android Studio, Vite, Docker

SELECTED AWARDS

Top 1% (8/940) Graduating Class GPA

May 2025

National Merit Scholar Finalist

February 2025

Congressional App Challenge Winner

December 2021

USACO Gold Division

December 2021